VR-Forms. Preprint.

Part V. Forms of Mathematics

Vitaly Reznik

2026

V.0. Introduction

Parts II–IV gave the apparatus of VR-Forms. Part V applies it to classical mathematics. The aim is to demonstrate how uncountable, paradoxical and non-operational mathematical objects find their place in the formal register, and how this changes their status.

Method of exposition: for each characteristic class of formal terms we give (1) the classical description, (2) status in the VR-Sets ontological register, (3) status in the formal register of VR-Forms, (4) methodological consequences.

Contents of Part V: classical ℝ as formal term (§V.1); classical ℘(ℕ) (§V.2); the Vitali set and AC for uncountable families (§V.3); the Russell term and paradoxical classes (§V.4); proper classes (NBG) and their status (§V.5); summary (§V.6).

V.1. Classical ℝ

Classical description

Classical ℝ — the set of all real numbers, defined in one of equivalent ways: as the complete ordered field, as the set of Dedekind cuts of ℚ, as the set of equivalence classes of fundamental sequences in ℚ. Cardinality $2^\wedge\aleph_0$ — uncountable.

Status in VR-Sets

In VR-Sets there is ℝ_VR — the operational set of computable reals, isomorphic to the field of computable reals (VR-Numbers §V.4). ℝ_VR is countable: each computable real is given by an algorithm, algorithms are countable.

The full classical ℝ is ontologically absent in VR-Sets. Not as «forbidden», but as a description signifying nothing: «the set of all reals» in the operational ontology does not specify a functionality (there is no procedure that would enumerate all reals).

Status in VR-Forms

ℝ is a formal term. Admissible in the formal register by the principle of forms. Not operationally realisable.

In the formal register one can formulate statements about ⌜ℝ⌝: «the cardinality of ⌜ℝ⌝ is $2^\wedge\aleph_0$», «ℝis complete as a metric space», « ℝis ordered». All these statements belong to the formal register. They do not generate ontological objects and have no ontological force.

Methodological consequences

Statements of the form «property P holds for all reals» in the formal register speak of ⌜ℝ⌝. To translate such a statement into the ontological register, one must show that for all elements of ℝ_VR (i.e. for all computable reals) the property P holds. This is passage via transit (§IV.2).

If P holds for all ℝ_VR — we obtain an ontological statement. If P holds for the classical ⌜ℝ⌝ but is not derivable for ℝ_VR without essential dependence on uncountability — this is the case of a **problematic transit** (§IV.4).

In practical mathematics most statements about ℝ relate to properties derivable from computability (completeness, ordering, field, metric). These statements transit directly. Statements essentially using uncountability (Liouville's theorem on transcendental numbers, the existence of measurable non-recursive sets) require separate analysis.

V.2. Classical ℘(ℕ)

Classical description

℘(ℕ) — the set of all subsets of ℕ. By Cantor's theorem it has cardinality 2^$\aleph_0$, uncountable.

Status in VR-Sets

℘_VR(ℕ) — the operational set of describable subsets of ℕ (VR-Sets §III.5). Countable. Contains all computable subsets of ℕ.

The statement «℘_VR(ℕ) is countable» is true in the ontological register. Cantor's diagonal argument in the operational ontology does not work for the same reasons as in Computable Analysis: there is no effective enumeration of all describable subsets (this is equivalent to the halting problem), so the diagonal construction is not operational.

Status in VR-Forms

⌜℘(ℕ)⌝ is a formal term. In the formal register the statement «⌜℘(ℕ)⌝ has cardinality 2^$\aleph_0$» is admissible. The statement «⌜℘(ℕ)⌝ is uncountable» is also admissible. Both belong to the formal register.

Here the difference of the two registers becomes visible. In the ontological: ℘_VR(ℕ) is countable. In the formal: ⌜℘(ℕ)⌝ is uncountable. No contradiction — these are statements about **different** objects. ℘_VR(ℕ) is an operational set; ⌜℘(ℕ)⌝ is a formal term; they are not identical.

Skolem's paradox in VR-Forms

In the classical context Skolem's paradox is the observation that ZFC has a countable model in which «℘(ℕ) is uncountable» is true (Löwenheim–Skolem). This is regarded as ontologically disconcerting: «how can ℘(ℕ) be countable from outside but uncountable from inside?»

In VR-Forms Skolem's paradox is removed explicitly. The «inside/outside» distinction becomes a distinction of registers. ℘_VR(ℕ) is countable ontologically («outside», in the full picture). $\ulcorner\wp(\mathbb{N})\urcorner$ is uncountable formally («inside», as a statement about a formal term). These are two different objects, each with its true status in its own register.

V.3. The Vitali set and AC for uncountable families

Classical description

The Vitali set — a choice of one representative from each equivalence class on $\mathbb{R}$ under the relation $x - y\ \mathbb{Q}$. The existence requires AC for an uncountable family (the number of equivalence classes is uncountable). Not Lebesgue measurable.

Status in VR-Sets

The Vitali set is ontologically absent in VR-Sets. The description «choose one from each class» is not operational: there is no procedure that would, on the request «give the next class», yield a definite representative (classes are uncountable). AC in VR-Sets is a theorem about countable families (VR-Sets §III.9 and §VI.4), not applicable to uncountable.

The Banach–Tarski paradox is absent in VR-Sets by construction, since it uses precisely Vitali-like uncountable choices (VR-Sets §VI.5).

Status in VR-Forms

$\ulcorner V\urcorner$ is the formal term for the Vitali set. In the formal register the statement «$\ulcorner V\urcorner$ exists» is admissible (by the principle of forms every description specifies a formal term). The statement «$\ulcorner V\urcorner$ is non-measurable» — formal.

⌜Banach–Tarski⌝ — a formal theorem about a paradoxical decomposition of ⌜the ball⌝ into a finite number of parts and assembly of two ⌜balls⌝. In the formal register admissible as a formal theorem. In the ontological register no analogue.

Methodological consequences

If in some context Vitali is used (for example, to construct a non-measurable set), this can be described in the formal register, but the operational correlate is absent. All applications essentially requiring Vitali remain in the formal register.

On the other hand, many classical results formally using AC for uncountable families can be rewritten without it — this is the programme of Reverse Mathematics. VR-Forms provides for this programme a natural frame: check whether the use of uncountable AC is essential (i.e. the transit is problematic in the sense of §IV.4) or eliminable (direct transit).

V.4. The Russell term and paradoxical classes

Classical description

Russell's class $R = \{x : x \notin x\}$. In naive set theory — source of the paradox: $R \in R \leftrightarrow R \notin R$. In ZF dissolved by the axiom of separation: one can form only $\{x \in A : x \notin x\}$ for an already existing A, and this does not generate R as a whole.

Status in VR-Sets

R is not an operational set (VR-Sets §II.3). The description «a functionality that yields x if and only if x does not yield itself» contains a contradiction on the query «$R \in R$?». The closure principle does not apply to contradictory descriptions.

Status in VR-Forms

$\ulcorner R \urcorner$ is a legitimate formal term. The principle of forms does not forbid contradictory descriptions (§II.4): syntactic correctness suffices.

In the formal register the statement $\ulcorner R \urcorner \in \ulcorner R \urcorner \leftrightarrow \ulcorner R \urcorner \notin \ulcorner R \urcorner$ is admissible. This is a formal statement about a formal term, derivable directly from the definition of $\ulcorner R \urcorner$.

This generates no contradiction in T_1. Proof: if from $\ulcorner R \urcorner \in \ulcorner R \urcorner \leftrightarrow \ulcorner R \urcorner \notin \ulcorner R \urcorner$ one could derive $\perp \in L_0$ (an ontological contradiction), then by conservativity $\perp$ would be derivable also in T_0, which is not the case.

A new reading of Russell

In the classical setting Russell's paradox is an indication of the necessity to restrict the formation of sets. In VR-Forms Russell's paradox has a different meaning: it shows that the description $\ulcorner R \urcorner$ **has no operational correlate**. This is not a defect of the description, but a property of it: some formal terms are operationally non-realisable.

Paradoxicality is a property of form, not an ontological failure. The liar, Berry, Grelling, Burali-Forti — all classical paradoxes in VR-Forms become statements about properties of formal terms, saying that the corresponding descriptions have no ontological correlate.

Remark on the freedom of the formal register

The admissibility of contradictory formal terms might seem alarming — but it does not break classical logic in the formal register. Logical operations apply to formulas as usual, and from a local contradiction (such as $\ulcorner R \urcorner \in \ulcorner R \urcorner \leftrightarrow \ulcorner R \urcorner \notin \ulcorner R \urcorner$) no global contradiction follows.

The point is that the contradiction $\ulcorner R \urcorner \in \ulcorner R \urcorner \leftrightarrow \ulcorner R \urcorner \notin \ulcorner R \urcorner$ is not a statement «is and is not at once»; it is a statement «the formal term $\ulcorner R \urcorner$ has such a property». The property characterises $\ulcorner R \urcorner$, separating it from operational sets. No universal logical catastrophe follows from it.

V.5. Proper classes

Classical description

In NBG (Bernays–Gödel) and MK (Morse–Kelley) extensions of ZFC, **proper classes** are introduced — collections too «large» to be sets. The class of all sets V, ordinals Ord, cardinals Card — proper classes. They are not elements of any set.

Status in VR-Sets

In VR-Sets proper classes are absent in the classical form, since the whole universe is countable (VR-Sets §VI.1) and the «too large» in the classical sense does not arise. However, there is an analogue: **describable families of functionalities that are not themselves**

functionalities. For instance, «the family of all computable functions $\mathbb{N} \to \mathbb{N}$» is a describable family but not an operational set (no procedure for their effective enumeration; this reduces to the halting problem).

Status in VR-Forms

$\ulcorner V \urcorner$ (the class of all sets) is a formal term. Admissible in the formal register. $\ulcorner \mathrm{Ord} \urcorner$, $\ulcorner \mathrm{Card} \urcorner$ — also formal terms.

The formal register plays here the same role as classes in NBG: it allows one to speak about large collections without introducing them into the ontology. Difference: in NBG classes are second-sort objects; in VR-Forms formal terms are formulas, not objects.

The hierarchy of t-applications

In VR-Sets §VII.3 the hierarchy of applications of the successor t was discussed as an operational analogue of Tarski's hierarchy of metalanguages. In VR-Forms this hierarchy receives an additional reading.

Each $t^n(\omega)$ is an operational set. Their union, «the entire hierarchy t^n», is a describable family that is not an operational set (it is an unbounded process). In the formal register it is a formal term $\ulcorner \bigcup_n t^n(\omega) \urcorner$ — an operational analogue of a proper class. This is an illustration: in VR-Forms the «too large» does not disappear; it passes into the formal register, remaining workable as a figure of speech.

V.6. Summary of Part V

Established:

(1) Classical $\mathbb{R}$ is a formal term $\ulcorner \mathbb{R} \urcorner$, not identical with the operational $\mathbb{R}_{VR}$ (§V.1).

(2) Classical $\wp(\mathbb{N})$ is a formal term $\ulcorner \wp(\mathbb{N}) \urcorner$, not identical with the operational $\wp_{VR}(\mathbb{N})$; Skolem's paradox is removed as a difference of registers (§V.2).

(3) The Vitali set and the Banach–Tarski paradox are formal terms; in the ontological register they are absent (§V.3).

(4) The Russell class is a legitimate formal term with a contradictory characteristic; paradoxicality is reinterpreted as a property of form indicating operational non-realisability (§V.4).

(5) Proper classes of NBG correspond to formal terms of VR-Forms; the hierarchy of t-applications provides the operational analogue of a proper class (§V.5).

General conclusion. The formal register of VR-Forms covers the whole classical set theory — uncountable collections, paradoxical classes, large classes of NBG — without placing any burden on the ontology beyond the countable operational universe. Classical mathematics becomes **speech** about formal terms, operationally realisable only in part; the programme of VR-Audit consists in a systematic classification of what in this speech has an ontological correlate and what does not.

What follows. Part VI extends the formal apparatus to non-mathematical descriptions — dragons, gods, philosophical categories — demonstrating the universality of the formal register. This is a natural completion of applications: the formal language of VR-Forms works with speech without exception, not only mathematical.